

input-output relations of an effector into a form approaching a simple proportionality. The feedback system, as we have seen, does more than this, and has a performance relatively independent of the characteristic and changes of characteristic of the effector used. The relative usefulness of the two methods of control thus depends on the constancy of the characteristic of the effector. It is natural to suppose that cases arise in which it is advantageous to combine the two methods. There are various ways of doing this. One of the most simple is that illustrated in the diagram of Fig. 4.

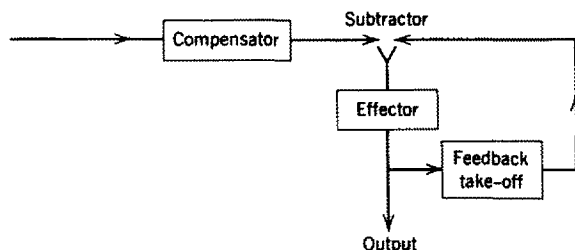


FIG. 4.

In this, the entire feedback system may be regarded as a larger effector, and no new point arises, except that the compensator must be arranged to compensate what is in some sense the average characteristic of the feedback system. Another type of arrangement is shown in Fig. 5.

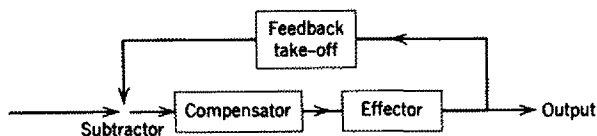


FIG. 5.

Here the compensator and effector are combined into one larger effector. This change will in general alter the maximum feedback admissible, and it is not easy to see how it can ordinarily be made to increase that level to an important extent. On the other hand, for the same feedback level, it will most definitely improve the performance of the system. If, for example, the effector has an essentially lagging characteristic, the compensator will be an anticipator or predictor, designed for its statistical ensemble of inputs. Our feedback, which we may call an anticipatory feedback, will tend to hurry up the action of the effector mechanism.